American Computer Science League

2023-2024 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$A\,(\,\overline{B\,C\;+\;\overline{\overline{A}}}\,)\;\;(\,\overline{\overline{\overline{A}}\,B\,C}\,)$$

A. 1
B. 3
C. 4
D. 5
E. 7

2. Data Structures

Find the internal path length of the binary search tree for:

OLDSMOBILECUTLASS

A. 41
B. 44
C. 46
D. 48
E. 54

3. Data Structures

There are 15 people who are in line for tickets to Taylor Swift's concert and there are 3 ticket booths. Customers are served in the order given at the first ticket booth available. The first 3 customers (Amy, Beth, and Cam) start at booths 1-3 in that order when the booths open.

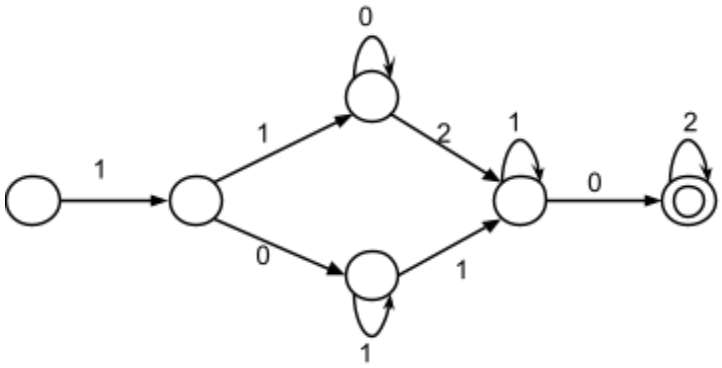
The times for each customer being at the ticket booth are as follows:
Amy = 15, Beth = 10, Cam = 8, Deb = 11, Eve = 4, Fran = 7, Gigi = 2, Hal = 12, Ivy = 9, Jan = 12, Kay = 6, Lexi = 8, Max = 8, Nara = 10, Omar = 5

Which booth finished last and how many minutes from the starting time did that booth finish serving its last customer?

A. Booth 1 - 42 min.
B. Booth 2 - 41 min.
C. Booth 3 - 44 min.
D. Booth 3 - 46 min.
E. Booth 2 - 45 min.

4. FSAs

Which of the following strings are accepted by this FSA?



a. 1101110 b. 1121022 c. 11021102

d. 1010 e. 1102112 f. 1011112

A. a, b, c, e
B. b, d, f
C. a, c, d
D. b, c, d
E. c, d, e, f

5. Regular Expressions

The ACSL Diner opened this year. The cook wants to order seasonings that satisfy the following regular expression. How many will he/she order?

[a-p]* e* [^s]* [d-hp-y]

relish	salt	mustard	pickle
pepper	ketchup	butter	vinegar
mayo	soy	salsa	honey

A. 4
B. 5
C. 6
D. 7
E. 8

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1. 布尔代数

请问**有多少个有序三元组可以使得下列布尔表达式的结果为真？**

$$A\,(\overline{B\,C\;+\;\overline{A}})\;\;(\overline{\overline{A}\,B\,C})$$

A. 1
B. 3
C. 4
D. 5
E. 7

2. 数据结构

求**以下字符串**构成**的二叉搜索树的内部路径长度：**

OLDSMOBILECUTLASS

A. 41
B. 44
C. 46
D. 48
E. 54

3. 数据结构

目前有 3 **个售票亭**，有 15 **个人**正在**排队购买泰勒斯威夫特演唱会门票。顾客们**按排队顺序，依次前往最先空闲出来的售票亭购票。**当**三个**售票亭**开始售票**时，前三位顾客**(Amy、Beth、Cam)按顺序在 1-3 **号售票亭**购票。

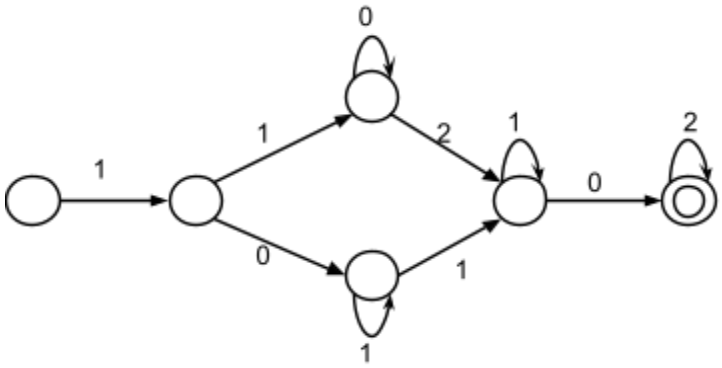
每位**顾客在购票亭**的**时间如下所示：**:
Amy = 15, Beth = 10, Cam = 8, Deb = 11, Eve = 4, Fran = 7, Gigi = 2, Hal = 12, Ivy = 9, Jan = 12, Kay = 6, Lexi = 8, Max = 8, Nara = 10, Omar = 5

请问**哪一个售票亭最后结束？这个售票亭从开门到服务完最后一位顾客一共使用多少分钟？**

A. 售票亭1 - 42 min.
B. 售票亭2 - 41 min.
C. 售票亭3 - 44 min.
D. 售票亭3 - 46 min.
E. 售票亭2 - 45 min.

4. 状态转换图

请问**下列哪些字符串可以被这个状态转换图**接受**？**



a. 1101110　　b. 1121022　　c. 11021102

d. 1010　　e. 1102112　　f. 1011112

A. a, b, c, e
B. b, d, f
C. a, c, d
D. b, c, d
E. c, d, e, f

5. 正则表达式

ACSL Diner 今年**开业。厨师想要订购一些满足以下正则表达式的调味料。请问他/她将订购多少种调味料？**

[a-p]* e* [^s]* [d-hp-y]

relish	salt	mustard	pickle
pepper	ketchup	butter	vinegar
mayo	soy	salsa	honey

A. 4
B. 5
C. 6
D. 7
E. 8